

Lab 1 - Local Home Lab Auditing

NETWORK DISCOVERY & ASSET INVENTORY

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1. Lab Overview

This lab establishes the first stage of a network security audit workflow: identify active assets, enumerate exposed TCP services, record addressing information, and convert the technical output into an initial asset inventory.

The environment is intentionally isolated and built with virtual machines. Kali Linux is used as the auditor, Windows 7 represents a legacy client endpoint, and Ubuntu Server represents a server workload. The VirtualBox host-only interface is also visible during discovery.

Lab topology

	IP address	Role
Kali Linux	10.10.10.10	Auditor / discovery host
Windows 7	10.10.10.20	Client endpoint
Ubuntu Server	10.10.10.30	Server
VirtualBox host-only interface	10.10.10.1	Host-side network interface

Addressing evidence

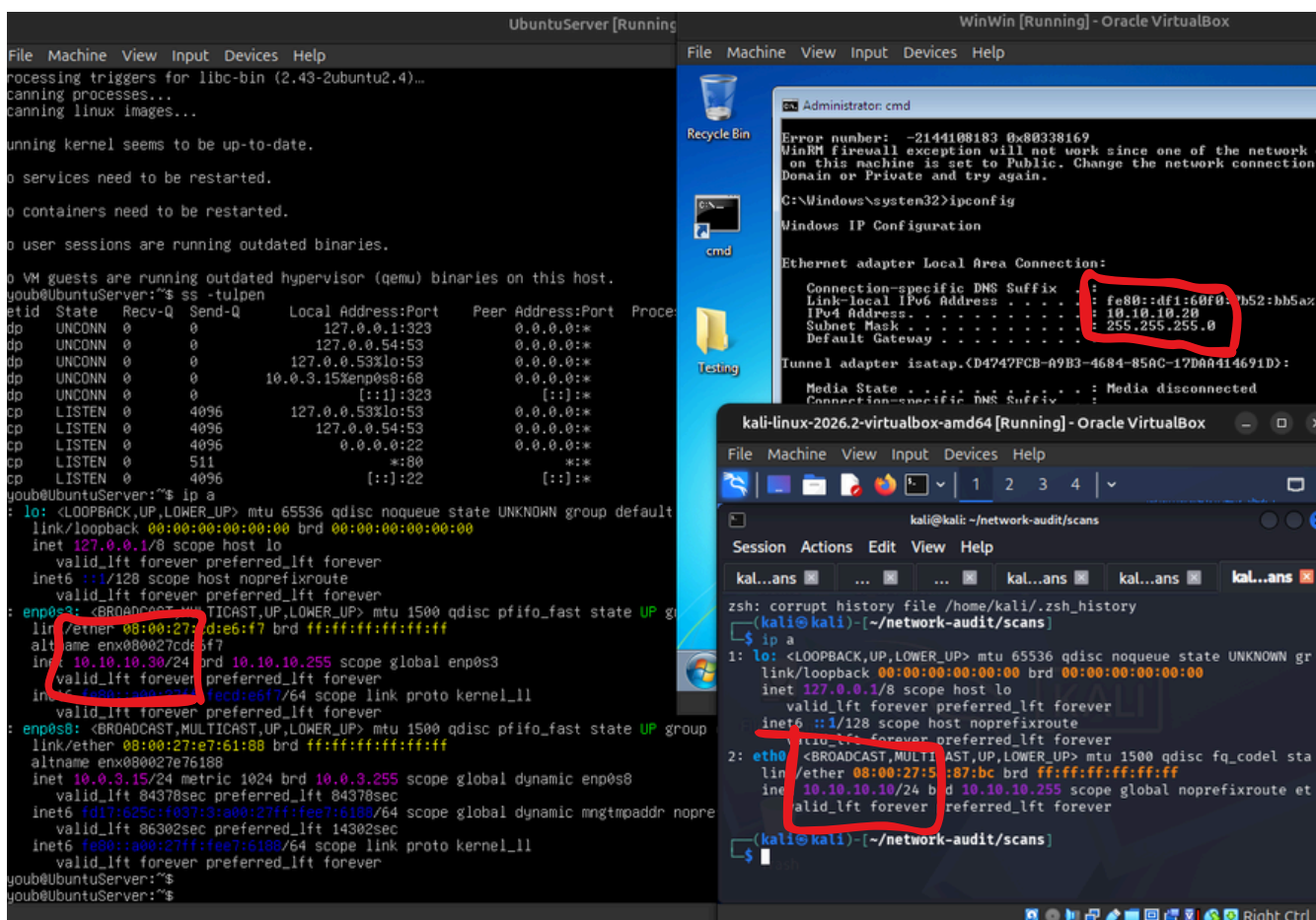


Figure 1. Lab addressing / topology evidence — placeholder for the original screenshot.

2. Discovery Methodology

The discovery scan was executed from Kali Linux against the isolated 10.10.10.0/27 range. The saved Nmap output records the exact command and scan scope.

```
/usr/lib/nmap/nmap -sS -sV -O -oA /home/kali/network-audit/scans/initial-discovery
10.10.10.0/27
```

Option summary

-sS	TCP SYN scan for identifying open TCP ports.
-sV	Service/version detection on discovered ports.
-O	Operating-system fingerprinting.
-oA	Save output in normal, greppable, and XML formats.
10.10.10.0/27	Scan 32 IPv4 addresses in the isolated lab range.

Kali scan evidence

```
Nmap scan report for 10.10.10.20
Host is up (0.0012s latency).
Not shown: 993 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
49153/tcp  open  msrpc        Microsoft Windows RPC
49154/tcp  open  msrpc        Microsoft Windows RPC
49155/tcp  open  msrpc        Microsoft Windows RPC
49156/tcp  open  msrpc        Microsoft Windows RPC
MAC Address: 08:00:27:F4:73:69 (Oracle VirtualBox virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: specialized|phone
Running: Microsoft Windows 7|Phone
OS CPE: cpe:/o:microsoft:windows_7 cpe:/o:microsoft:windows
OS details: Microsoft Windows Embedded Standard 7, Microsoft Windows Phone 7.5 or 8.0
Network Distance: 1 hop
Service Info: Host: WINWIN; OS: Windows; CPE: cpe:/o:microsoft:windows

Nmap scan report for 10.10.10.30
Host is up (0.0012s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 10.2p1 Ubuntu 2ubuntu3.6 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http         Apache httpd 2.4.66 ((Ubuntu))
MAC Address: 08:00:27:CD:E6:F7 (Oracle VirtualBox virtual NIC)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.99%E=4%D=9/27%OT=22%CT=1%CU=41143%PV=Y%DS=1%DC=D%G=Y%M=080027%T
OS:M=6AB94F60%P=x86_64-pc-linux-gnu)SEQ(SP=100%GCD=1%ISR=100%TI=Z%CI=Z%II=I
OS:%TS=21)SEQ(SP=101%GCD=1%ISR=103%TI=Z%CI=Z%II=I%TS=22)SEQ(SP=103%GCD=1%IS
OS:R=100%TI=Z%CI=Z%TS=21)SEQ(SP=105%GCD=1%ISR=10C%TI=Z%CI=Z%II=I%TS=21)SEQ
OS:SP=106%GCD=1%ISR=10E%TI=Z%CI=Z%II=I%TS=21)OPS(O1=M5B4ST11NW8%O2=M5B4ST11
OS:NW8%O3=M5B4NNT11NW8%O4=M5B4ST11NW8%O5=M5B4ST11NW8%O6=M5B4ST11)WIN(W1=FE8
OS:8%W2=FE88%W3=FE88%W4=FE88%W5=FE88%W6=FE88)ECN(R=Y%DF=Y%T=40%W=FAF0%O=M5B
OS:4NNSNW8%CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=0%A=S+%F=AS%RD=0%Q=)T2(R=N)T3(R=N)T4(
OS:R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%F
OS:=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T7(R=Y%DF=Y%T
OS:=40%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=6%RI
OS:D=G%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40%CD=S)
```

Figure 2. Kali discovery scan evidence — placeholder for the original Kali screenshot.

3. Discovery Results

The saved Nmap output reports four responsive addresses: 10.10.10.1, 10.10.10.20, 10.10.10.30, and the Kali scanner at 10.10.10.10. The scan completed across 32 addresses.

IP	Observed role / host	Observed services	Key observation
10.10.10.1	VirtualBox host-only interface	No open TCP ports in scanned set	MAC identified; OS fingerprint not specific.
10.10.10.10	Kali / scanner	No open TCP ports in scanned set	Local scanner; network distance 0 hops.
10.10.10.20	WINWIN / Windows 7 client	135, 139, 445, 49153–49156/tcp	RPC, NetBIOS and SMB-related exposure identified.
10.10.10.30	Ubuntu server	22/tcp SSH; 80/tcp HTTP	OpenSSH 10.2p1 and Apache httpd 2.4.66 detected.

Windows 7 enumeration

Nmap identified TCP/135 (MSRPC), TCP/139 (NetBIOS Session Service), TCP/445 (Microsoft-DS/SMB), and TCP/49153–49156. The host was reported as WINWIN and associated with a Windows 7 CPE. Nmap also warned that the OS result may be unreliable because of the port-state mix, so the fingerprint should be treated as an enumeration result rather than an absolute determination.

Ubuntu server enumeration

Nmap identified TCP/22 running OpenSSH 10.2p1 for Ubuntu and TCP/80 running Apache httpd 2.4.66. The scan did not produce an exact OS match, but its service information identified the host as Linux.

4. Asset Inventory

The following table reproduces the supplied CSV inventory. The CSV records the initial role and authorization status of each planned asset; the Nmap results are then used to enrich the observed service information.

IP	Host	OS	Role	Services	Status
10.10.10.10	Kali	Kali Linux	Auditor	—	Authorized
10.10.10.20	WIN7-CLIENT	Windows 7	Client	TBD	Authorized
10.10.10.30	UBUNTU-SRV	Ubuntu	Server	SSH/HTTP	Authorized

Inventory reconciliation

The supplied CSV initially marked the Windows client services as **TBD**. The completed Nmap scan resolves that field with observed TCP services. For Ubuntu Server, the CSV already recorded SSH/HTTP and the Nmap results independently confirmed TCP/22 and TCP/80.

This separation between planned inventory and observed network exposure is useful in an audit workflow: the inventory states what is expected, while discovery establishes what is actually reachable.

5. Local Service Verification

The supplied Ubuntu Server screenshot provides local corroboration of the services exposed by the server. The displayed **ss -tulpen** output shows listeners corresponding to Apache on TCP/80 and SSH on TCP/22, matching the Nmap observations.

```
ayoub@UbuntuServer:~$ ss -tulpen
Netid State  Recv-Q Send-Q   Local Address:Port   Peer Address:Port   Process
udp    UNCONN 0      0      127.0.0.1:323        0.0.0.0:*           ino:12250 sk:1 cgroup:/system.slice/chrony.service <->
udp    UNCONN 0      0      127.0.0.54:53        0.0.0.0:*           uid:990 ino:6048 sk:2 cgroup:/system.slice/systemd-resolved.service <->
udp    UNCONN 0      0      127.0.0.53%lo:53     0.0.0.0:*           uid:990 ino:6046 sk:3 cgroup:/system.slice/systemd-resolved.service <->
udp    UNCONN 0      0      10.0.3.15%enp0s8:68  0.0.0.0:*           uid:990 ino:12107 sk:4 cgroup:/system.slice/systemd-networkd.service <->
udp    UNCONN 0      0      [::]:323            [::]:*             ino:12251 sk:5 cgroup:/system.slice/chrony.service v6only:1 <->
tcp    LISTEN 0      4096    127.0.0.53%lo:53     0.0.0.0:*           uid:990 ino:6047 sk:6 cgroup:/system.slice/systemd-resolved.service <->
tcp    LISTEN 0      4096    127.0.0.54:53        0.0.0.0:*           uid:990 ino:6049 sk:7 cgroup:/system.slice/systemd-resolved.service <->
tcp    LISTEN 0      4096    0.0.0.0:22          0.0.0.0:*           ino:19769 sk:8 cgroup:/system.slice/ssh.socket <->
tcp    LISTEN 0      511      *:80                *:*                 ino:19439 sk:9 cgroup:/system.slice/apache2.service v6only:0 <->
tcp    LISTEN 0      4096    [::]:22            [::]:*             ino:19771 sk:a cgroup:/system.slice/ssh.socket v6only:1 <->
```

Figure 3. Ubuntu Server local listening-socket verification supplied with the lab.

The screenshot also shows local UDP listeners associated with system services. These are not treated as externally exposed TCP findings in this initial report because the Nmap assessment documented the TCP discovery scope used for this portfolio exercise.

6. Reproducibility Notes

The scan artifacts should be kept with the project repository so the result can be reproduced and compared with later audit passes. The XML output is useful for future automation, while the normal and grepable outputs provide convenient human-readable and processing-oriented formats.

Next lab preparation

This report intentionally remains limited in scope. It documents the initial discovery phase rather than presenting a complete security assessment. The next labs can build on the same isolated topology by introducing more realistic services, configuration checks, vulnerability assessment, and remediation validation.

The practical outcome is a repeatable baseline: identify assets, enumerate services, preserve evidence, reconcile observations with an asset inventory, and document the work in a form suitable for a technical portfolio.

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